

Statistics

2025 Lectures

Part 2 - Random Variables

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A numerical function of the outcomes

- In most cases, an experimenter focuses on specific characteristics of the experiment.
- For example, a traffic engineer may concentrate on the number of vehicles traveling on a particular road or in a specific direction, rather than on the brand of vehicles or the number of passengers in each vehicle.
- Each outcome of the experiment can be associated with a number by establishing a rule of association.
- The transition from experimental outcomes to a numerical function of these outcomes is facilitated by the concept of a random variable, a useful tool for describing events and much more.

Random variable

- A general idea: a random variable is a number that depends on chance.
- The aim is to define a random variable that is associated with the phenomenon being studied, rather than with a specific sample space.

Definition 1: A real-valued function $X[(S, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}^1)]$ is a **random variable**, provided it is **measurable**, i.e.

$$B \in \mathcal{B}^1 \Rightarrow X^{-1}(B) \in \mathcal{A}.$$

- Hence, each $\{s \in S | X(s) \leq t\}$ or for brevity just $\{X \leq t\}$ is an event for each $t \in \mathbb{R}$.

Intuition: Think of X as the information you care about encoded as a number: monthly inflation rate, unemployment indicator, log-return, time-to-job-finding.

Types of RVs

- **Discrete**: takes countable values (e.g., $0, 1, 2, \dots$).
Examples: number of defaults in a quarter; unemployment status (0/1).
- **Continuous**: takes values on intervals of $\mathbb{R}$. Examples: inflation π_t , log-returns r_t , income (often modeled as lognormal).
- **Mixed**: mixture of discrete mass and continuous density (e.g., spike at zero consumption change plus continuous part).

Invariance of random variables

Definition 2: X, X' random variables, defined on sample spaces S, S' , respectively, describing the same phenomenon, are **equivalent** if for every t we have $\{X \leq t\}$ occurs if and only if $\{X' \leq t\}$ occurs and these events have the same probability.

- the above equivalence satisfies the requirements for relation of equivalence - reflexivity, symmetry and transitivity

Example 1: In the tossing of three fair coins, let the random variable X be defined as the number of tails. The possible values for X are 0, 1, 2, 3. We can associate probabilities with these values as follows:

$$P(X = 0) = P(\{H, H, H\}) = 1/8;$$

$$P(X = 1) = P(\{H, H, T\} \cup \{H, T, H\} \cup \{T, H, H\}) = 3/8;$$

$$P(X = 2) = P(\{H, T, T\} \cup \{T, H, T\} \cup \{T, T, H\}) = 3/8;$$

$$P(X = 3) = P(\{T, T, T\}) = 1/8.$$

Distribution of random variable

Definition 3: By **distribution** of the random variable X we mean the assignment of probabilities to all events $\{X \in A\} \subset S$, where $A \subset \mathbb{R}$.

- Basic type of events are given by intervals

$$\{a < X < b\}, \{a \leq X < b\}, \{a < X \leq b\}, \{a \leq X \leq b\},$$

for $-\infty \leq a \leq b \leq \infty$.

Discrete distribution

Definition 4: A random variable X is called **discrete** if there is a finite or countable set of real numbers $U = \{x_1, x_2, \dots\}$ such that

$$P(X \in U) = \sum_n P(X = x_n) = 1.$$

Experiment	Random Variable (x)	Possible Values for the Random Variable
Contact five customers	Number of customers who place an order	0, 1, 2, 3, 4, 5
Inspect a shipment of 50 radios	Number of defective radios	0, 1, 2, $\dots$, 49, 50
Operate a restaurant for one day	Number of customers	0, 1, 2, 3, $\dots$
Sell an automobile	Gender of the customer	0 if male; 1 if female

Definition 5: The **probability mass function** (or discrete density function) of a discrete random variable X is the function

$$p_X(k) = \begin{cases} P(X = k) & \text{if } k \in U; \\ 0 & \text{else.} \end{cases}$$

Discrete workhorses

- Certain distributions with specific characteristics have special names.
- There are many classes of parametric distributions for random variables.
- **Uniform** $\{a, b\}$: $X \in \{x_1, \dots, x_n\}$, $P(X = x_i) = 1/n$.
Random allocation of scarce resources, e.g. housing units, a random draw among all applicants.
- **Bernoulli** (p) : $X \in \{0, 1\}$, $P(X = 1) = p$. Indicator of unemployment, policy take-up.
- **Binomial** (n, p) : Sum of n independent Bernoullis.
Number of adopters in a sample.
- **Poisson** (λ) : Counts of arrivals in time/space. Defaults per month, calls, accidents.

Uniform distribution

Example 2: $U = \{x_1, \dots, x_n\}$, $P(X = x_i) = 1/n, i = 1, \dots, n$.

For example, when selecting a winning lottery ticket from a total of n tickets (assuming each ticket has an equal chance of winning).

Such distribution of the random variable X is called a **discrete uniform distribution**.

Binomial distribution

Example 3:

$$U = \{0, 1, \dots, n\}, P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, k = 0, \dots, n$$

A random variable with this distribution is often defined as the total number of successes in n independent experiments, each with a probability p of success.

The probability of obtaining k successes and $n - k$ failures in any specific order equals $p^k (1 - p)^{n-k}$. There are $\binom{n}{k}$ different ways to arrange these outcomes. Clearly,

$$\sum_{k=0}^n P(X = k) = \sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} = (p + (1 - p))^n = 1.$$

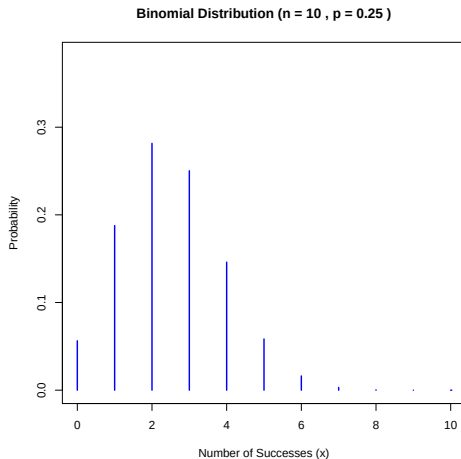
This type of distribution is called a **binomial** distribution.

We write $X \sim \text{BIN}(n, p)$.

Binomial distribution: Example

Example 4: Consider $X \sim \text{BIN}(10, 0.25)$.

The graph of the **probability mass function** (pmf)



More examples of discrete distribution

Example 5:

$$U = \{0, 1, 2, \dots\}, P(X = k) = p(1 - p)^k, k = 0, 1, 2, \dots$$

Random variable X represents the number of trials before the first success in a series.

$$\sum_{k=0}^{\infty} P(X = k) = p \sum_{k=0}^{\infty} (1 - p)^k = p \frac{1}{p} = 1.$$

This type of distribution is called a **geometric** distribution and we denote it as $X \sim GEO(p)$.

Example 6:

$$U = \{0, 1, 2, \dots\}, P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, k = 0, 1, 2, \dots$$

This type of distribution is called a **Poisson** distribution.

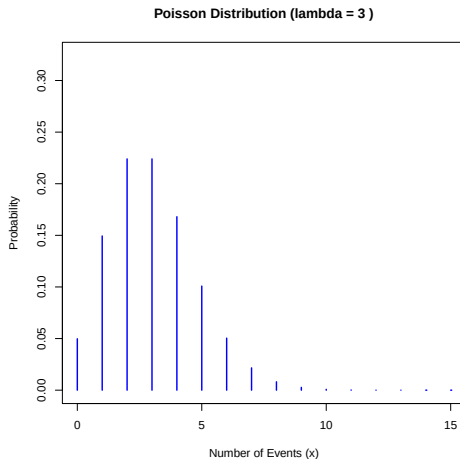
$$\sum_{k=0}^{\infty} P(X = k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$$

We write $X \sim POI(\lambda)$.

Poisson distribution: Example

Example 7: Consider $X \sim POI(3)$.

The graph of its **probability mass function** (pmf)



Random variable and events

Theorem 1: The probabilities of the events of the form $\{a < X \leq b\}$ for all $-\infty \leq a \leq b \leq \infty$ uniquely determine the probabilities of events of the form $\{a < X < b\}$, $\{a \leq X < b\}$ and $\{a \leq X \leq b\}$.

- Moreover $\{a < X \leq b\} = \{X \leq b\} \setminus \{X \leq a\}$ and so

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a).$$

- Thus, the probabilities of events $\{a < X \leq b\}$ are determined by the probabilities of events $\{X \leq t\}$ for $-\infty < t < \infty$.

Distribution function

Definition 6: For any random variable X , the (cumulative) **distribution function** (cdf) F_X is defined as

$$F_X(t) = P(X \leq t), \quad t \in \mathbb{R}.$$

Theorem 2: For any random variable X , cdf F_X has the following properties:

- a) F_X is nondecreasing;
 - b) $\lim_{t \rightarrow -\infty} F_X(t) = 0$, $\lim_{t \rightarrow \infty} F_X(t) = 1$;
 - c) F_X is continuous on the right.
- As a consequence $P(X = t) = F_X(t) - F_X(t)_-$. So if F_X is continuous at t , then $P(X = t) = 0$.
- For discrete random variables,

$$F_X(t) = P(X \leq t) = \sum_{x_i \leq t} P(X = x_i).$$

Properties of cdf: Example

Example 1 (cont.): Recall that X represents the number of tails in three tosses of a fair coin. In such simple cases, the distribution can be represented using a table.

values	0	1	2	3
probabilities	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

We have $0 \leq X \leq 3$. So $F_X(t) = 0$ for $t < 0$; $F_X(t) = 1$ for $t \geq 3$.

For $t \in [0, 1)$

$$F_X(t) = P(X = 0) = 1/8$$

For $t \in [1, 2)$

$$F_X(t) = P(X = 0) + P(X = 1) = 1/2$$

For $t \in [2, 3)$

$$F_X(t) = P(X = 0) + P(X = 1) + P(X = 2) = 7/8$$

Remark: The cumulative distribution function (cdf) is defined for all real values, not just for the values of the random variable. For example, one can compute $F_X(2.27)$.

Continuous random variable

Definition 7: The random variable X is called continuous if there exists a nonnegative function f , called the **probability density function** (pdf) of X , such that

$$F_X(t) = \int_{-\infty}^t f(x) \mathrm{d}x, \quad -\infty < t < \infty.$$

Theorem 2c: For a continuous random variable X ,

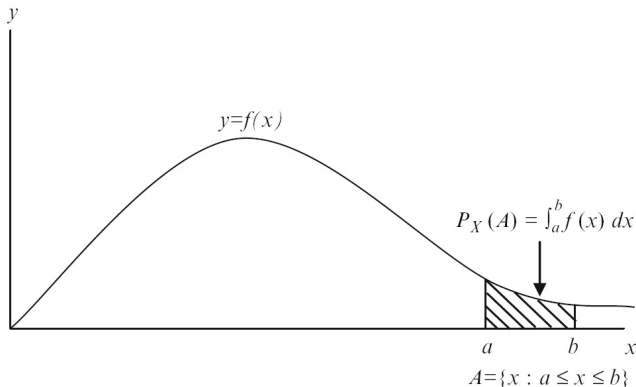
- a) F_X is nondecreasing;
- b) $\lim_{t \rightarrow -\infty} F_X(t) = 0$, $\lim_{t \rightarrow \infty} F_X(t) = 1$;
- c) F_X is continuous.

- As a consequence $P(X = t) = 0$ for any $t \in \mathbb{R}$.
- $f(t) = F'(t)$ and $f(t) \geq 0$ almost everywhere

Properties of cdf for continuous distribution

Theorem 3: If X has density f then for all $a < b$ we have

$$\begin{aligned} P(a \leq X \leq b) &= P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) = \\ &= F(b) - F(a) = \int_a^b f(x) dx. \end{aligned}$$



Continuous workhorses

- **Uniform**(a, b): simple baseline on $[a, b]$.
- **Exponential**(λ): memoryless durations; constant hazard $h(x) = \lambda$.
- **Normal**(μ, σ^2): measurement error, aggregation, CLT preview.
- **Lognormal**(μ, σ^2): positive, right-skewed quantities (income, expenditures).

Uniform distribution

Example 8: Let $a < b$ and

$$f(x) = \begin{cases} 0, & \text{if } X < a; \\ c, & \text{if } a \leq X \leq b; \\ 0, & \text{if } X > b. \end{cases}$$

Then,

$$F_X(t) = \begin{cases} 0, & \text{if } t < a; \\ \int_a^t f(x) \mathrm{d}x = c(t - a), & \text{if } a \leq t \leq b; \\ c(b - a), & \text{if } t > b. \end{cases}$$

Hence, $c := \frac{1}{b-a}$ and the distribution of X is called (continuous) **uniform** on the interval $[a, b]$.

We denote this as $X \sim U[a, b]$.

Exponential distribution

Example 9: The distribution with the probability density function (pdf)

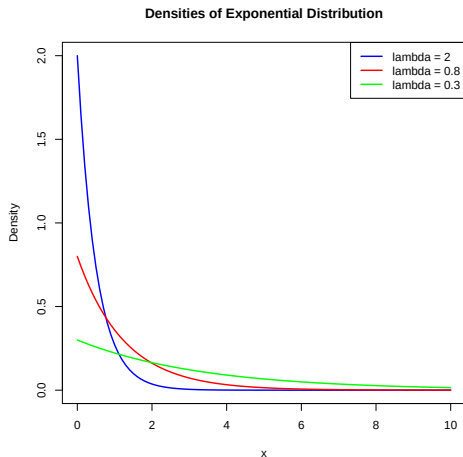
$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0; \\ 0, & x \leq 0, \end{cases}$$

is called an **exponential** distribution with parameter $\lambda > 0$. We denote this as $X \sim EXP(\lambda)$.

- Job-finding or inter-trade times
- Rate λ governs both tail and pace.

Exponential distribution: Example

Example 10: Consider $X \sim EXP(\lambda)$ for values $\lambda = 0.3, 0.8, 2$.
The graph of the corresponding density functions

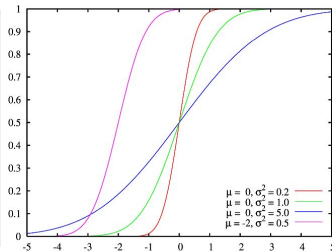
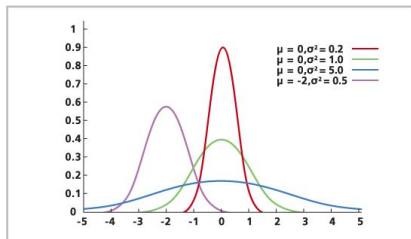


Normal distribution

Example 11: The distribution with the probability density function (pdf)

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad x \in \mathbb{R},$$

is called a **normal** distribution with parameters μ and σ^2 . We denote this as $X \sim N(\mu, \sigma^2)$.



Standard normal distribution

For any $a < b$, the probability $P(a \leq X \leq b)$ is given by

$$P(a \leq X \leq b) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{z^2}{2}} dz = \Phi(z_2) - \Phi(z_1),$$

where $z_1 = \frac{a-\mu}{\sigma}$ and $z_2 = \frac{b-\mu}{\sigma}$. Here, Φ is the cumulative distribution function of the **standard normal distribution**, denoted as $N(0, 1)$.

Moreover,

$$\Phi(-z) = 1 - \Phi(z).$$

Quantiles

Definition 8: Let X be a random variable with cdf F_X , and let $0 < p < 1$. The p th (lower) **quantile** ξ_p of X is defined as a minimum of solutions of inequalities

$$P(X \leq x) \geq p, \quad P(X \geq x) \geq 1 - p.$$

- Alternatively, the p th quantile ξ_p has to satisfy

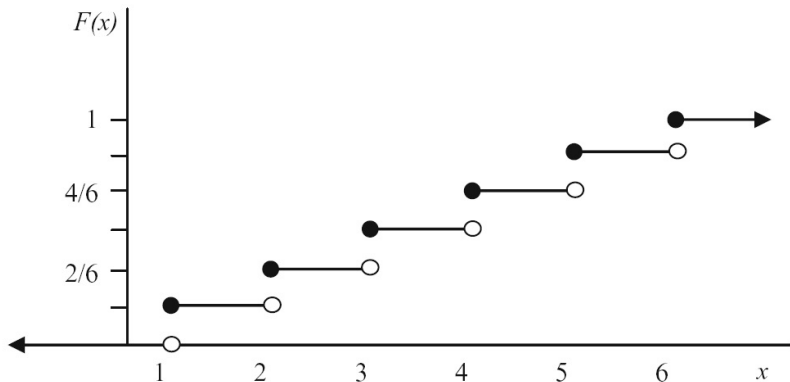
$$F_X(\xi_p) \geq p \text{ and } F_X(\xi_p)_- \leq p.$$

- for $p = 0.25$, ξ_p is called the lower quartile
- for $p = 0.5$, ξ_p is called the median
- for $p = 0.75$, ξ_p is called the upper quartile
- alternatively, p th **upper quantile** is the $(1 - p)$ th lower quantile

Quantiles: Example

Example 12: Consider rolling a die.

The graph of the cumulative distribution function (cdf) would show the cumulative probabilities for each outcome (1 to 6).



Find median and upper and lower quartiles.

More properties of continuous random variables

- Let X be a continuous random variable. Then the p th quantile ξ_p of X is defined as a minimum value of $x \in \mathbb{R}$ satisfying

$$P(X \leq x) = p, 0 < p < 1.$$

Theorem 4: Let X be a random variable with continuous cdf F_X . Then

$$P(\xi_a \leq X \leq \xi_b) = b - a \quad \text{for } 0 < a < b < 1.$$

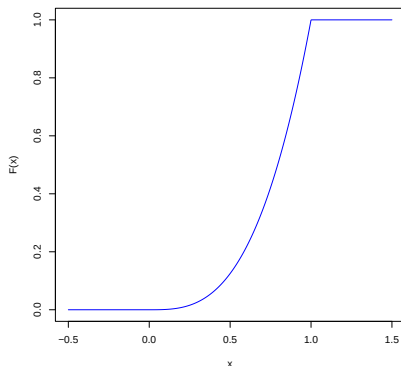
Example 13: If F is continuous, there is always a 50% probability that a random variable with cdf F will assume a value between its lower quartile (Q1) and upper quartile (Q3). This interval, known as the interquartile range (IQR), encompasses the middle 50% of the data.

Example 14

Example 14: Determine the median, lower and upper quartiles for random variable X with the following cdf, $k > 0$:

$$F_X(x) = \begin{cases} 0, & \text{for } x < 0; \\ kx^3, & \text{for } 0 \leq x \leq \frac{1}{\sqrt[3]{k}}; \\ 1, & \text{for } x > \frac{1}{\sqrt[3]{k}}. \end{cases}$$

Example 14: cdf for $k=1$



More properties of cdfs

Theorem 5: For any function F satisfying a) - c) (Theorem 2) there is a probability space $(S, \mathcal{A}, P)$ and a random variable X such that $F_X = F$.

- Note that different random variables can have the same cdf.

Example 15: Let us consider the experiment consisting of three tosses of a coin, along with two random variables: X , representing the total number of heads, and Y , representing the total number of tails.

A straightforward count reveals that $P(X = k)$ and $P(Y = k)$ are the same for all values of k . Consequently, the cumulative distribution functions F_X and F_Y are identical.

Transformation of random variable

- In practical situations, we often work with a function (transformation) of a random variable.

Example 16: A computer can generate a random number from a uniform distribution on the interval $[0, 1]$. If the desired random number needs to be from a different range $[a, b]$, $a < b$, one simply needs to apply a linear transformation

$$\varphi(x) = (b - a)x + a.$$

- Functions of random variables are very important in statistics.

Transformation of discrete random variables

- Let X take values in the set $U = \{x_1, x_2, \dots\}$ with corresponding probabilities $p_i = P(X = x_i)$, such that $\sum p_i = 1$. Then, if $Y = \varphi(X)$, where φ is a real-valued function, Y will also have a discrete distribution.
- Since the function φ may not be one-to-one, we can express the probability as follows

$$P(Y = y) = P(\varphi(X) = y) = \sum_{x: \varphi(x)=y} P(X = x).$$

Example 17: Suppose that X has the following distribution

values	-2	-1	0	1	2	3	4
probabilities	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$

If we define the transformation $\varphi(x) = x^2$, $x \in \mathbb{R}$, then the random variable $Y = X^2$ will have the following distribution

values	0	1	4	9	16
probabilities	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{2}{10}$

Transformation of continuous random variables

- Let F and f represent the cdf and pdf of the continuous random variable X , respectively. Consider a transformation $Y = \varphi(X)$, where φ is assumed to be at least piecewise differentiable.
- First, we will examine the case where φ is strictly monotone.
- If φ is strictly increasing, then

$$F_Y(y) = P(\varphi(X) \leq y) = P(X \leq \varphi^{-1}(y)) = F_X(\varphi^{-1}(y)).$$

- If φ is strictly decreasing, then

$$F_Y(y) = P(\varphi(X) \leq y) = P(X \geq \varphi^{-1}(y)) = 1 - F_X(\varphi^{-1}(y)).$$

Theorem 6: If φ is a continuous differentiable function with inverse ψ and X is a continuous random variable with density f_X , then the density of $Y = \varphi(X)$ is

$$f_Y(y) = f_X(\psi(y))|\psi'(y)|, y \in \mathbb{R}.$$

Probability integral transformation

Example 18: Let $Y = aX + b$. If $a > 0$, then for $y \in \mathbb{R}$,

$$F_Y(y) = P(aX + b \leq y) = P\left(X \leq \frac{y - b}{a}\right) = F_X\left(\frac{y - b}{a}\right)$$

and

$$f_Y(y) = \frac{1}{a} \cdot f_X\left(\frac{y - b}{a}\right).$$

Theorem 7: (Probability integral transformation)

Let X be a continuous random variable with strictly increasing cdf F . Then the distribution of $Y = F(X)$ is uniform on $[0, 1]$.

PIT histograms: calibration check

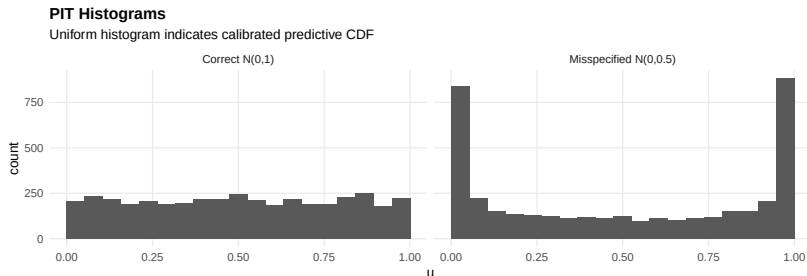


Figure: Illustration: Uniform histogram $\Rightarrow$ well-calibrated predictive CDF; U-shape or slope indicates mis-specification.

For example: Evaluate density forecasts (inflation/FX), Uniformity is the diagnostic target.

Transformation of a continuous random variable

- When φ is not monotone, we still have

$$F_Y(y) = P(\varphi(X) \leq y).$$

However, since φ does not have an inverse function, the inequality $\varphi(X) \leq y$ typically does not correspond to a single inequality for X . In this case, the analysis becomes more complex, as we must consider the intervals where X meets the condition $\varphi(X) \leq y$.

Example 19: Let X be a continuous random variable and define $\varphi(x) = x^2, x \in \mathbb{R}$. Then, we have $F_Y(y) = 0$ for $y \leq 0$, and for $y > 0$

$$F_Y(y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y}).$$

Also, for $y > 0$

$$f_Y(y) = F'_Y(y) = \frac{1}{2\sqrt{y}}(f_X(\sqrt{y}) + f_X(-\sqrt{y})).$$